

Supplementary Examination – Summer 2022

Course :T.Y. B.Tech.

Branch : Civil Engg.

Semester : V

Subject Code & Name : BTCVC503 Soil Mechanics

Max Marks : 60

Date : / /

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the Following		
A) Explain the terms voids ratio, specific gravity and Bulk density.	CO1	06
B) Enlist the methods for determination of water content. Explain any one.	CO1	06
C) Derive relation between bulk unit weight, specific gravity, voids ratio and degree of saturation.	CO1	06
Q.2 Solve Any Two of the Following		
A) Explain Textural classification of Soil.	CO1	06
B) Explain derivation of two dimensional flow through Laplace equation.	CO2	06
C) Explain graphical method for determination of Pheratic line through earthen dam.	CO2	06
Q.3 Solve Any Two of the Following		
A) Explain Vane shear test with neat sketch.	CO2	06
B) Explain procedure of determination of maximum dry density by standard proctor test.	CO3	06
C) What is vertical stress? State at least four assumption made by Boussinesq's equation.	CO2	06
Q.4 Solve Any two of the Following		
A) Explain theory of compaction and factors influencing compaction.	CO2	06
B) Explain construction and use of New mark influence chart.	CO3	06
C) What are the factors contributing slope failure?	CO3	06
Q.5 Solve Any Two of the Following		
A) Explain the Mohr-coulomb's theory of failure.	CO3	06
B) Explain stress isobar with help of sketch.	CO3	06
C) Derive the expression for active pressure assuming backfill as dry.	CO3	06

